EXPT NO:8	FORECASTING AND PREDICTIVE ANALYTICS USING TABLEAU AND POWER BI
DATE: 14.08.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is forecasting in the context of Business Intelligence (BI)?

2. Explain the difference between predictive analytics and descriptive analytics.

3. Name two common forecasting methods used in Tableau and Power BI.

4. Why is data granularity important for accurate forecasting?

5. Name two challenges in forecasting using multiple data sources.

IN-LAB EXERCISE:

OBJECTIVE:
To perform time series forecasting and predictive analytics on historical sales data using Tableau and Power BI, and visualize future trends for informed decision-making.
PROCEDURE:

1. Data Collection:

- Import historical sales data from multiple sources:
  - Dataset 1: Sales Data (CSV) → Order ID, Order Date, Region, Sales Amount
  - Dataset 2: Promotions Data (Excel) → Promotion ID, Region, Promotion Start Date, Promotion End Date, Discount

2. Data Preparation and Transformation:

- Clean data: standardize date formats, remove duplicates, handle missing values.
- Aggregate sales data at the required granularity (daily, weekly, or monthly).
- Ensure numeric fields (Sales, Discount) are correct.

3. Data Integration and Modeling:

- **Tableau:** Connect datasets and relate them using the Region field.
- **Power BI:** Create relationships in the Model view; define cardinality where necessary.

4. Forecasting and Predictive Analytics:

- **Tableau:**
  - Use the Forecast feature on Sales over time.
  - Apply trend models like Exponential Smoothing and visualize prediction intervals.
- **Power BI:**
  - Use built-in forecasting in line charts.
  - Optionally, create a DAX measure for predicted sales using moving averages or linear regression.

5. Create Calculated Fields:

- Forecasted Sales = Predicted value from the model
- Sales Growth % = ((Forecasted Sales – Actual Sales) / Actual Sales) * 100

6. Design a Comprehensive Dashboard:

- Line chart showing actual vs. forecasted sales over time.
- KPI cards: Total Forecasted Sales, Expected Growth %.
- Use combo charts to show sales trends along with promotion periods.
- Highlight regions with expected growth above or below target thresholds.

7. Validate Forecast Accuracy:

- Compare forecasts against actual historical sales (back-testing).
- Adjust forecasting parameters if necessary (e.g., seasonality length, smoothing factor).

8. Publishing and Sharing:

- **Tableau:** Publish workbook to Tableau Public or Tableau Server.
- **Power BI:** Publish report to Power BI Service and share with stakeholders.

Experiment Datasets:

- **Dataset 1:** Historical Sales Data (CSV) → Order ID, Order Date, Region, Sales
- **Dataset 2:** Promotions Data (Excel) → Promotion ID, Region, Promotion Start/End Date, Discount

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. How did forecasting help in predicting future sales trends for business planning?

2. What forecasting method did you use, and why was it suitable for the dataset?

3. What challenges, if any, did you face regarding missing data or irregular time intervals?

4. How can predictive analytics influence decision-making in a retail environment?

5. What additional data (e.g., competitor pricing, economic indicators) could be integrated to improve forecast accuracy?

RESULT:

The historical sales and promotions datasets were successfully cleaned, integrated, and modeled in Tableau and Power BI. Forecasting techniques provided predicted sales trends, growth percentages, and region-wise performance expectations. The resulting dashboards allowed managers to visualize actual vs. forecasted sales, enabling better planning and data-driven decision-making.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		